

Assignment No. 6

Subject: Database Engineering Lab (7CS271)

Class: SY CSE, AY 2025–26, Sem-II

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Batch - S5

PART A: AGGREGATE FUNCTIONS & GROUP ANALYSIS

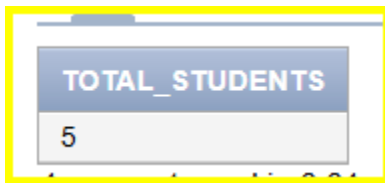
1. Summary Statistics

a. Count total records

Query

```
SELECT COUNT(*) AS Total_Students FROM Student;
```

Result



TOTAL_STUDENTS
5

b. Average CGPA

Query

```
SELECT AVG(cgpa) AS Avg_CGPA FROM Student;
```

Result

AVG_CGPA
8.2

c. Minimum & Maximum CGPA

Query

```
SELECT MIN(cgpa) AS Min_CGPA, MAX(cgpa) AS Max_CGPA FROM Student;
```

Result

MIN_CGPA	MAX_CGPA
7.5	9

d. Total credits

Query

```
SELECT SUM(credits) AS Total_Credits FROM Course;
```

Result

TOTAL_CREDITS
18

2. Group-Based Analysis

a. Count students per semester

Query

```
SELECT semester, COUNT(*) AS Student_Count  
  
FROM Student  
  
GROUP BY semester;
```

Result

SEMESTER	STUDENT_COUNT
6	2
4	2
8	1

b. Total students per course

Query

```
SELECT course_id, COUNT(*) AS Total_Students  
  
FROM Enrolls  
  
GROUP BY course_id;
```

Result

COURSE_ID	TOTAL_STUDENTS
504	1
502	1
505	1
503	1
501	1

c. Average CGPA per semester

Query

```
SELECT semester, AVG(cgpa) AS Avg_CGPA  
FROM Student  
GROUP BY semester;
```

Result

SEMESTER	AVG_CGPA
6	8
4	8.75
8	7.5

3. HAVING (Group Filtering)

a. Semesters with more than 1 student

Query

```
SELECT semester, COUNT(*) AS Count  
FROM Student  
GROUP BY semester  
HAVING COUNT(*) > 1;
```

Result

SEMESTER	COUNT
6	2
4	2

b. Semesters with avg CGPA > 8

Query

```
SELECT semester, AVG(cgpa) AS Avg_CGPA  
FROM Student  
GROUP BY semester  
HAVING AVG(cgpa) > 8;
```

Result

SEMESTER	AVG_CGPA
4	8.75

c. Courses with more than 1 enrollment

Query

```
SELECT course_id, COUNT(*) AS Total  
FROM Enrolls  
GROUP BY course_id  
HAVING COUNT(*) > 1;
```

Result

no data found

PART B: SUBQUERIES

4. Single Row Subqueries

a. Students above average CGPA

Query

```
SELECT * FROM Student
```

```
WHERE cgpa > (SELECT AVG(cgpa) FROM Student);
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
101	1	07/01/2022	-	4	8.5
102	2	07/01/2022	-	4	9

2 rows returned in 0.00 seconds Download

b. Students with maximum CGPA

Query

```
SELECT * FROM Student
```

```
WHERE cgpa = (SELECT MAX(cgpa) FROM Student);
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
102	2	07/01/2022	-	4	9

1 rows returned in 0.00 seconds Download

5. Multi-Row Subqueries

a. Students enrolled in courses

Query

```
SELECT * FROM Student
```

```
WHERE student_id IN (SELECT student_id FROM Enrolls);
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
101	1	07/01/2022	-	4	8.5
102	2	07/01/2022	-	4	9
103	3	07/01/2021	-	6	7.8
104	4	07/01/2021	-	6	8.2
105	5	07/01/2020	-	8	7.5

b. Persons who are students

Query

```
SELECT * FROM Person
```

```
WHERE person_id IN (SELECT person_id FROM Student);
```

Result

PERSON_ID	FIRST_NAME	LAST_NAME	STREET_NO	STREET_NAME	CITY	ZIP_CODE	DATE_OF_BIRTH	EMAIL
1	Aarav	Sharma	12	MG Road	Pune	411001	05/10/2004	a1@gmail.com
2	Diya	Patel	45	FC Road	Pune	411004	08/15/2003	a2@gmail.com
3	Rohan	Mehta	22	Link Road	Mumbai	400050	01/20/1990	a3@gmail.com
4	Sneha	Iyer	18	Brigade Road	Bangalore	560001	03/12/1985	a4@gmail.com
5	Karan	Singh	10	Civil Lines	Delhi	110001	11/25/1992	a5@gmail.com

6. Correlated Subqueries

a. Students above semester average

Query

```
SELECT s.student_id, s.cgpa, s.semester  
FROM Student s  
WHERE cgpa > (  
    SELECT AVG(cgpa)  
    FROM Student  
    WHERE semester = s.semester  
);
```

Result

STUDENT_ID	CGPA	SEMESTER
102	9	4
104	8.2	6

b. Faculty who teach at least one course

Query

```
SELECT * FROM Faculty f
WHERE EXISTS (
SELECT 1 FROM Teaches t
WHERE t.faculty_id = f.faculty_id
);
```

Result

FACULTY_ID	PERSON_ID	DESIGNATION	OFFICE_LOCATION
201	6	Professor	CS Block
202	7	Associate Professor	IT Block
203	8	Assistant Professor	AI Block
204	9	Lecturer	DS Block
205	10	Professor	ML Block

Faculty who teach at least one course

Faculty who teach at least one course

PART C: SET OPERATIONS

7. UNION

a. Combine student & faculty person IDs

Query

```
SELECT person_id FROM Student
```

```
UNION
```

```
SELECT person_id FROM Faculty;
```

Result

PERSON_ID
1
2
3
4
5
6
7
8
9
10

b. Removes duplicates automatically

8. UNION ALL

a. Combine with duplicates

Query

```
SELECT person_id FROM Student
```

```
UNION ALL
```

```
SELECT person_id FROM Faculty;
```

Result

PERSON_ID
3
1
2
4
5
8
9
6
7
10

b. Difference:

- UNION → removes duplicates
- UNION ALL → keeps duplicates

9. INTERSECT

a. Common person IDs

Query

```
SELECT person_id FROM Student  
  
INTERSECT  
  
SELECT person_id FROM Faculty;
```

Result

no data found

10. MINUS

a. Students who are not faculty

Query

```
-- SELECT person_id FROM Student  
  
MINUS  
  
SELECT person_id FROM Faculty;
```

Result

PERSON_ID
1
2
3
4
5

b. Students not teaching courses

Query

```
SELECT student_id FROM Student  
  
MINUS  
  
SELECT student_id FROM Enrolls;
```

Result

PERSON_ID
1
2
3
4
5